

ZIJIE LI

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🌐 [Personal Website](#)

PROFESSIONAL EXPERIENCE

Research Scientist on media generation,

Working on text-to-image generation and multi-modal unified generation

ByteDance Seed *San Jose, CA*

April, 2025 - Now

Research intern on multimodal diffusion models,

Turning pretrained text-to-image diffusion model into a unified image-text diffusion model [[paper](#)]

Mentor: [Linjie Yang](#), [Peng Wang](#)

ByteDance Seed *San Jose, CA*

May, 2024 - November, 2024

EDUCATION

Ph.D. in Mechanical Engineering,

Topic: AI for Science, focusing on neural networks for simulation across different scales

Advisor: [Amir Barati Farimani](#)

Carnegie Mellon University, GPA: 4.00/4.00

January, 2020 - March, 2025

B.E. in Theoretical and Applied Mechanics,

Sun Yat-sen University, GPA: 3.91/4.00

August, 2015 - May, 2019

RESEARCH INTEREST

Numerical simulation of PDEs and dynamical systems, Physics and numerical methods inspired design of neural networks, Diffusion models, Generative AI, Vision-language models

SELECTED PEER-REVIEWED PUBLICATIONS

(For the up-to-date publication list please refer to the [Google scholar](#), * denotes equal contribution)

1. Dual Diffusion for Unified Image Generation and Understanding [[Paper](#)], [[Code coming soon](#)]
Conference on Computer Vision and Pattern Recognition 2025
Zijie Li*, Henry Li*, Yichun Shi, Yuval Kluger, A. Barati Farimani, Linjie Yang, Peng Wang
2. Scalable Transformer for PDE surrogate modelling [[Paper](#)], [[Code](#)]
Advances on Neural Information Processing Systems 2023
Zijie Li, Dule Shu, A. Barati Farimani
3. Transformer for Partial Differential Equations' Operator Learning [[Paper](#)], [[Code](#)]
Transactions on Machine Learning Research (2023)
Zijie Li, Kazem Meidani, A. Barati Farimani
4. A physics-informed diffusion model for high-fidelity flow field reconstruction [[Paper](#)], [[Code](#)]
Journal of Computational Physics (2023)
Dule Shu*, **Zijie Li***, A. Barati Farimani
5. Graph Neural Network Accelerated Molecular Dynamics [[Paper](#)], [[Code](#)]
Journal of Chemical Physics (2022)
Zijie Li, Kazem Meidani, Prakarsh Yadav, A. Barati Farimani

6. TPU-GAN: Learning temporal coherence from dynamic point cloud sequences [Paper], [Code]
International Conference on Learning Representations 2022
Zijie Li, Tianqin Li, A. Barati Farimani
7. Prototype memory and attention mechanisms for few shot image generation [Paper], [Code]
International Conference on Learning Representations 2022
 Tianqin Li*, **Zijie Li***, Andrew Luo, Harold Rockwell, A. Barati Farimani, Tai Sing Lee
8. Graph neural network-accelerated Lagrangian fluid simulation [Paper], [Code]
Computers & Graphics (2022)
Zijie Li, A. Barati Farimani

CONFERENCE PRESENTATION

1. **Latent Neural PDE Solver for time-dependent system**
 37th Annual Conference on Neural Information Processing Systems, AI for Science workshop
 New Orleans, LA December 2023
2. **Factorized kernel attention for scalable PDE learning**
 76th Annual Meeting of the Division of Fluid Dynamics, Washington DC November, 2023
3. **Mesh-agnostic PDE Operator Learning with Attention**
 American Physical Society (APS) March 2023, Las Vegas, NV March, 2023
4. **Accelerating Lagrangian fluid simulation with graph neural networks**
 International Conference on Learning Representations 2021 SimDL workshop, Virtual May, 2021
5. **Graph Neural Network for Lagrangian Fluid Simulation**
 73th Annual Meeting of the Division of Fluid Dynamics, Virtual November, 2020

INDUSTRIAL COLLABORATION

- Transformer-based neural operator** (with Nvidia) September, 2023 - May, 2024
 Contributing attention-based kernel integral and Transformer utilities to the open-source library: **neuraloperator**.
- Physics-informed diffusion model** (with Nvidia) October, 2023 - March, 2024
 Contributing physics-informed diffusion to Nvidia's physics+machine learning library: **Modulus**.
- Neural operator for reaction-diffusion simulation** (with KLA Tencor) February, 2022 - December, 2022
 Created a differentiable simulation pipeline for simulating reaction-diffusion data and studied different kinds of neural operator surrogates with physics-informed loss.

BOOK CHAPTERS

- Graph Neural Networks for Molecules
 A chapter for book "Machine Learning in Molecular Sciences" published by Springer Nature
 Yuyang Wang, **Zijie Li**, Amir Barati Farimani

TEACHING EXPERIENCE

- Teaching Assistant September, 2023 - December, 2023
 24889 (Online certificate course): Deep learning for engineers
- Teaching Assistant January, 2023 - May, 2023
 24788: Introduction to Deep Learning & 24789: Intermediate to Deep Learning

SERVICE

Reviewer:

Conference: NeurIPS, ICLR, ICML, CVPR

Journal: Nature Machine Intelligence, IEEE Transactions on Neural Networks and Learning Systems, Transaction on Machine Learning Research

SKILLS

Programming: Python, Fortran, C++

Package: PyTorch, Deep Graph Library, Pytorch Geometric, Numba, Phiflow, OpenMM, JAX, JAX-MD, FENICS

Language: Mandarin (native), English (proficient)

SELECTED COURSEWORKS

Deep reinforcement learning and control, Numerical methods, Computer vision, Introduction to machine learning, Introduction to deep learning, Learning-based image synthesis, Engineering computation (C++), Computational fluid dynamics, Finite element analysis, Molecular simulation for materials